COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structures and Algorithms	**SECTION:**	CPE21S1
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ACTIVITY NO. 5.1 **Queues**

1. PROCEDURE OUTPUT

ILO A: Create queue using C++ STL

Source file

```
// ILO A: Create queue using C++ STL

#include <iostream>
#include <queue>

//prototype
void display(std::queue<char> r);

int main(){

    std::queue<char> q;

    //push
    q.push('c'); // front
    q.push('p');
    q.push('e');
    q.push('0');
    q.push('1');
    q.push('0'); // rear



    //empty()
    std::cout<<"Is the queue empty: " << q.empty()<<std::endl;
```

```
    //size
    std::cout<<"Size of the queue: "<< q.size()<<std::endl;

    //front
    std::cout<< "Front of the queue: "<<q.front()<<std::endl; // front

    //back
    std::cout<<"Back of the queue: "<<q.back()<<std::endl;

    //pop
    std::cout<<"Pop: ";
    q.pop();

    //push another element
    std::cout<< "Push another element: ";
    q.push('b');

    //dislpay
    std::cout<<"Displaying element/s: ";
    display(q);

    //delete the queue
    while(!q.empty()){
        q.pop();
    }

}


void display(std::queue<char> r){
    // crete a copy of the queue
    std::queue<char> c = r;

    while(!c.empty()){
        std::cout << " "<<c.front();
        c.pop();
```

```
    }
    std::cout<<"\n";
}
```

Header source file

```
#ifndef QUEUE_H
#define QUEUE_H
#include <iostream>

template <typename T>
class qNode{
    public:
    T data;
    qNode *next;

};

//creating a new node.
template  <typename T>
qNode<T>*  new_node(T newdata){
    qNode<T>* newNode = new qNode<T>;
    newNode->data = newdata;
    newNode->next = nullptr;
    return newNode;

}

template<typename T>
void enqueue (qNode<T>** frontPtr, qNode<T>** backPtr, T newData){
    //creating a NEW NODE
    qNode<T>* newNode = new_node(newData);

    //INSERTING TO AN EMPTY QUEUE
    if((*frontPtr) == nullptr && (*backPtr)==nullptr){
        (*frontPtr) = newNode;
```

```
        (*backPtr) = newNode;
    }
    //inserting an item into a non empt queue
    else {
        // point the backPtr NEXT TO THE newNode
        (*backPtr)->next = newNode;
        (*backPtr) = newNode;
    }
}

template<typename T>
void dequeue (qNode<T>** frontPtr, qNode<T>** backPtr){
    //check if the queue is empty to prevent crashes
    if((*frontPtr) == nullptr){
        return;
    }

    //createa temporary node to store the node to be deleted
    qNode<T>* deleteNode = nullptr;
    deleteNode = (*frontPtr);

    //check if the queue is only 1 node
    if((*frontPtr) == (*backPtr)){
        (*frontPtr) = nullptr;
        (*backPtr) = nullptr;
        delete deleteNode;
        return;
    }

    //deleting of the node
    (*frontPtr) = deleteNode->next;
    delete deleteNode;

    }

    //display all elements in the list
template<typename T>
```

```
void display (qNode<T>* frontPtr){
    qNode<T> *temp = frontPtr;

    while(temp !=nullptr){
        std::cout<< " " << temp->data;
        temp = temp->next;
    }
}

    //return the front
template<typename T>
T front(qNode<T>* frontPtr){
    return frontPtr->data;
}

#endif
```

OUTPUT

```
Is the queue empty: 0
Size of the queue: 6
Front of the queue: c
Back of the queue: 0
Pop: Push another element: Displaying element/s:  p e 0 1 0 b
```

Code analysis:

This file tests C++'s built-in std:: queue. It pushes 6 characters in, then runs through every basic queue operation, it checks if it's empty, checking size, peeking at front/back, popping one off, pushing one more, then displaying everything.

B.1. Linked List Implementation

Source file

```cpp
#include <iostream>
#include "1queuell.h"

int main(){
    qNode<char>* front = nullptr;
    qNode<char>* back = nullptr;

    enqueue(&front, &back, 'J');
    std::cout << front->data << " " << back->data << std::endl;

    enqueue(&front, &back, 'I');
    std::cout << front->data << " " << back->data << std::endl;

    enqueue(&front, &back, 'M');
    std::cout << front->data << " " << back->data << std::endl;

    dequeue(&front, &back);
    std::cout << front->data << " " << back->data << std::endl;

    return 0;
}
```

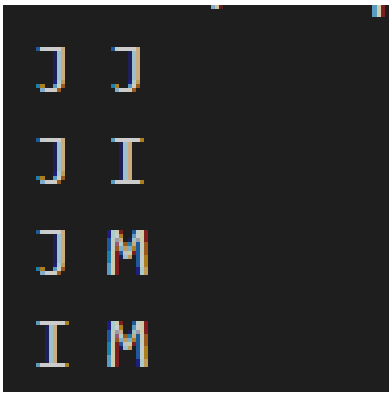
Header source file

```cpp
#ifndef QUEUE_H
#define QUEUE_H
#include <iostream>

template <typename T>
class qNode {
public:
    T data;
    qNode *next;
};

// creating a new node
```

```cpp
template <typename T>
qNode<T>* new_node(T newdata) {
    qNode<T>* newNode = new qNode<T>;
    newNode->data = newdata;
    newNode->next = nullptr;
    return newNode;
}

template<typename T>
void enqueue(qNode<T>** frontPtr, qNode<T>** backPtr, T newData) {
    // creating a NEW NODE
    qNode<T>* newNode = new_node(newData);

    // INSERTING TO AN EMPTY QUEUE
    if((*frontPtr) == nullptr && (*backPtr) == nullptr) {
        (*frontPtr) = newNode;
        (*backPtr) = newNode;
    }
    // inserting an item into a non empty queue
    else {
        // point the backPtr NEXT TO THE newNode
        (*backPtr)->next = newNode;
        (*backPtr) = newNode;
    }
}

template<typename T>
void dequeue(qNode<T>** frontPtr, qNode<T>** backPtr) {
    // check if the queue is empty to prevent crashes
    if((*frontPtr) == nullptr) {
        return;
    }

    // create a temporary node to store the node to be deleted
    qNode<T>* deleteNode = nullptr;
    deleteNode = (*frontPtr);
```

```
    // check if the queue is only 1 node
    if((*frontPtr) == (*backPtr)) {
        (*frontPtr) = nullptr;
        (*backPtr) = nullptr;
        delete deleteNode;
        return;
    }

    // deleting of the node
    (*frontPtr) = deleteNode->next;
    delete deleteNode;
}

// display all elements in the list
template<typename T>
void display(qNode<T>* frontPtr) {
    qNode<T> *temp = frontPtr;
    while(temp != nullptr) {
        std::cout << " " << temp->data;
        temp = temp->next;
    }
}

// return the front
template<typename T>
T front(qNode<T>* frontPtr) {
    return frontPtr->data;
}

#endif
```

OUTPUT



Code analysis:

This code builds a custom queue from scratch using linked nodes. It manually manages memory pointers to add items at the end and remove items from the front.

B.2. Array-Based Implementation

Source file

```
// PART B.2: Array-Based (Circular) Queue Implementation

#include <iostream>
#include "2queuearr.h"

int main() {
    queueArr<int> myQueue(5);   // a queue that can hold 5 numbers

    std::cout << "Enqueue 10, 20, 30, 40, 50\n";
    myQueue.Enqueue(10);
    myQueue.Enqueue(20);
    myQueue.Enqueue(30);
    myQueue.Enqueue(40);
    myQueue.Enqueue(50);

    std::cout << "\nFront: " << myQueue.Front() << "\n";
```

```
    std::cout << "Back : " << myQueue.Back() << "\n";
    std::cout << "Size : " << myQueue.Size() << "\n";
    std::cout << "Full?: " << std::boolalpha << myQueue.Full() << "\n";

    std::cout << "\nDequeue twice\n";
    myQueue.Dequeue();
    myQueue.Dequeue();
    std::cout << "Front now: " << myQueue.Front() << "\nSize: " <<
myQueue.Size() << "\n";

    std::cout << "\nEnqueue 60, 70\n";
    myQueue.Enqueue(60);
    myQueue.Enqueue(70);
    std::cout << "Back now: " << myQueue.Back() << "\nFull?: " <<
myQueue.Full() << "\n";

    std::cout << "\nCopy constructor:\n";
    queueArr<int> copy(myQueue);
    std::cout << "\ncopy.Front(): " << copy.Front() << "\ncopy.Back():
" << copy.Back() << "\n";

    std::cout << "\nClear()\n";
    myQueue.Clear();
    std::cout << "Empty? " << myQueue.Empty() << "\n";

    return 0;
}
```

Header source file

```
#ifndef QUEUEARR_H
#define QUEUEARR_H

#include <cstddef>    // gives us size_t
#include <stdexcept>  // gives us std::runtime_error
```

```cpp
template <typename T>
class queueArr {
private:
    T* data;            // the array that holds the items
    size_t capacity;     // how many items the array can hold
    size_t frontIndex;   // index of the front item
    size_t backIndex;    // index of the back item
    size_t count;        // how many items are currently stored

public:
    bool Empty();
    bool Full();
    size_t Size();
    void Clear();

    T Front();
    T Back();

    void Enqueue(T newData);
    void Dequeue();

    queueArr(size_t cap);
    ~queueArr();

    queueArr(const queueArr& other);              // copy constructor
    queueArr& operator=(const queueArr& other);    // copy assignment
};

// Build a new, empty queue that can hold "cap" items.
template <typename T>
queueArr<T>::queueArr(size_t cap) {
    capacity = cap;
    count = 0;
    frontIndex = 0;
    backIndex = 0;
    data = new T[capacity];
}
```

```cpp
// Free the array when the queue is destroyed.
template <typename T>
queueArr<T>::~queueArr() {
    delete[] data;
}

// Make a brand-new queue that is a copy of "other".
template <typename T>
queueArr<T>::queueArr(const queueArr& other) {
    capacity = other.capacity;
    frontIndex = other.frontIndex;
    backIndex = other.backIndex;
    count = other.count;

    data = new T[capacity];
    for (size_t i = 0; i < capacity; i++) {
        data[i] = other.data[i];
    }
}

// Copy the contents of "other" into a queue that already exists.
template <typename T>
queueArr<T>& queueArr<T>::operator=(const queueArr& other) {
    if (this == &other) {
        return *this;
    }

    delete[] data;   // free our old array first

    capacity = other.capacity;
    frontIndex = other.frontIndex;
    backIndex = other.backIndex;
    count = other.count;

    data = new T[capacity];
    for (size_t i = 0; i < capacity; i++) {
```

```
        data[i] = other.data[i];
    }

    return *this;
}

// Is the queue empty?
template <typename T>
bool queueArr<T>::Empty() {
    return count == 0;
}

// Is the queue full?
template <typename T>
bool queueArr<T>::Full() {
    return count == capacity;
}

// How many items are stored rn?
template <typename T>
size_t queueArr<T>::Size() {
    return count;
}

// Reset the queue back to empty.
template <typename T>
void queueArr<T>::Clear() {
    frontIndex = 0;
    backIndex = 0;
    count = 0;
}

// front of the queue.
template <typename T>
T queueArr<T>::Front() {
    if (Empty()) {
        throw std::runtime_error("Front() called on empty queue");
```

```
    }
    return data[frontIndex];
}

// back of the queue.
template <typename T>
T queueArr<T>::Back() {
    if (Empty()) {
        throw std::runtime_error("Back() called on empty queue");
    }
    return data[backIndex];
}

// Add new item to the back of the queue.
template <typename T>
void queueArr<T>::Enqueue(T newItem) {
    if (Full()) {
        throw std::runtime_error("Enqueue() called on full queue");
    }

    if (count == 0) {
        backIndex = 0;   // this is the very first item
    } else {
        // move one step forward, wrapping back to 0 after the last
slot
        backIndex = (backIndex + 1) % capacity;
    }

    data[backIndex] = newItem;
    count++;
}

// Remove item at the front of the queue.
template <typename T>
void queueArr<T>::Dequeue() {
    if (Empty()) {
        throw std::runtime_error("Dequeue() called on empty queue");
```

```
    }
    frontIndex = (frontIndex + 1) % capacity;    // wrap around if
needed
    count--;
}

#endif
```

OUTPUT

```
Enqueue 10, 20, 30, 40, 50

Front: 10
Back : 50
Size : 5
Full?: true

Dequeue twice
Front now: 30
Size: 3

Enqueue 60, 70
Back now: 70
Full?: true

Copy constructor:

copy.Front(): 30
copy.Back(): 70

Clear()
Empty? true
```

Code analysis:

This code creates a fixed-size circular queue using a standard array. It uses index wrapping to efficiently re-use array slots as items are added and removed.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

ILO C: Solve problems using queue implementation

Source file

```
#include <iostream>

class Job {
public:
    int id;
    std::string u;
    int p;

    Job() {
        id = 0;
        u = "";
        p = 0;
    }

    Job(int a, std::string b, int c) {
        id = a;
        u = b;
        p = c;
    }
};

struct JobNode {
    Job data;
    JobNode* next;

    JobNode(Job j) : data(j), next(nullptr) {}
};
```

```
class JobQueue {
public:
    JobNode* f;
    JobNode* b;
    int c;

    JobQueue() {
        f = nullptr;
        b = nullptr;
        c = 0;
    }

    bool isEmpty() {
        if (f == nullptr) return true;
        else return false;
    }

    int size() {
        return c;
    }

    void enqueue(Job j) {
        JobNode* n = new JobNode(j);
        if (f == nullptr) {
            f = n;
            b = n;
        } else {
            b->next = n;
            b = n;
        }
        c = c + 1;
    }

    bool dequeue(Job &o) {
        if (f == nullptr) return false;
        JobNode* t = f;
        o = t->data;
```

```
        f = f->next;
        if (f == nullptr) b = nullptr;
        delete t;
        c = c - 1;
        return true;
    }

    ~JobQueue() {
        Job x;
        while (dequeue(x)) {}
    }
};

class Printer {
public:
    JobQueue q;
    int nid;

    Printer() {
        nid = 1;
    }

    void addJob(std::string u, int p) {
        Job j(nid, u, p);
        nid++;
        q.enqueue(j);
        std::cout << "Added job #" << j.id << " for " << u << " (" << p
<< " pages) to the list." << std::endl;
    }

    void processAllJobs() {
        std::cout << std::endl << "Printer starting! Total jobs to
print: " << q.size() << std::endl;
        Job cur;
        while (q.dequeue(cur)) {
            std::cout << "Printing job #" << cur.id << " for " << cur.u
<< " (" << cur.p << " pages)... Done!" << std::endl;
```

```
        }
        std::cout << "All done! No jobs left." << std::endl;
    }

    bool hasPendingJobs() {
        if (!q.isEmpty()) return true;
        else return false;
    }
};

int main() {
    Printer p;

    std::cout << "People are adding print jobs" << std::endl;
    p.addJob("User 1", 3);
    p.addJob("User 2", 12);
    p.addJob("User 3", 1);
    p.addJob("User 4", 7);

    p.processAllJobs();

    std::cout << std::endl << "New jobs came in!" << std::endl;
    p.addJob("User 6", 5);
    p.addJob("User 7", 2);

    p.processAllJobs();

    return 0;
}
```

OUTPUT

```
People are adding print jobs
Added job #1 for User 1 (3 pages) to the list.
Added job #2 for User 2 (12 pages) to the list.
Added job #3 for User 3 (1 pages) to the list.
Added job #4 for User 4 (7 pages) to the list.

Printer starting! Total jobs to print: 4
Printing job #1 for User 1 (3 pages)... Done!
Printing job #2 for User 2 (12 pages)... Done!
Printing job #3 for User 3 (1 pages)... Done!
Printing job #4 for User 4 (7 pages)... Done!
All done! No jobs left.

New jobs came in!
Added job #5 for User 6 (5 pages) to the list.
Added job #6 for User 7 (2 pages) to the list.

Printer starting! Total jobs to print: 2
Printing job #5 for User 6 (5 pages)... Done!
Printing job #6 for User 7 (2 pages)... Done!
All done! No jobs left.
```

Code analysis:

This code simulates a real-world print job manager using a linked-list queue. It processes document requests one by one in the exact order they were received.

Conclusion:

Queues are fundamental first-in, first-out **(FIFO)** data structures that can be created using C++ STL, custom linked lists, or circular arrays. Learning these different approaches demonstrates how queues efficiently organize sequential tasks, such as real-world printer job management.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic

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work. The following indicates the nature and extent of AI assistance:

Tools Used: Claude

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: *Idea generation, formatting, summarization, code assistance, and explanation of cncepts*

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: I used AI as a helper to make my explanations easy to read. I wrote the code logic, tested it .

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.